

Schwarz–Christoffel Maps onto Periodic Comb Domains

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What is a conformal map?

A conformal map on the complex plane is angle-preserving in the following sense:

Definition

An analytic map, $f : U \rightarrow \mathbb{C}$ where U is an open subset of $\mathbb{C}$, is said to be *conformal* if the angles between any two intersecting curves in U are preserved under f .

Definition

Two domains, $A, B \subseteq \mathbb{C}$ are said to be *conformally equivalent* if there exists a conformal map $f : A \rightarrow B$, that is, a holomorphic bijection between A and B .

Theorem

A function $f : U \rightarrow \mathbb{C}$ is a conformal mapping onto its image $f(U)$ if and only if it is holomorphic and injective on U .

Canonical domains

Definition

The *unit disk* is denoted by $\mathbb{D}$ and is defined as $\mathbb{D} := \{z \in \mathbb{C} : |z| < 1\}$.

Definition

The *upper half-plane* is denoted by $\mathbb{H}$ and is defined as $\mathbb{H} := \{z \in \mathbb{C} : \Im z > 0\}$

Theorem

The upper half-plane and the unit disk are conformally equivalent. In particular, let $f(z) := \frac{i - z}{i + z}$. Then $f : \mathbb{H} \rightarrow \mathbb{D}$ is a conformal map with inverse

$$f^{-1}(w) = i \frac{1 - w}{1 + w}$$

The bijective holomorphisms from the unit disk onto itself are the *automorphisms* of the unit disk, and assume a particular form; $f(z) = e^{i\theta} \frac{z - a}{1 - \bar{a}z}$

Schwarz–Christoffel Maps

A *Schwarz–Christoffel* map is a type of conformal map where the target domain is a polygon in the complex plane. Typically, the domain is the upper half-plane or the unit disk, although other domains such as a strip of the complex plane can be used.

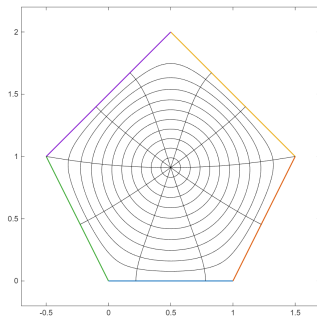


Figure 1: Map from unit disk onto pentagon.

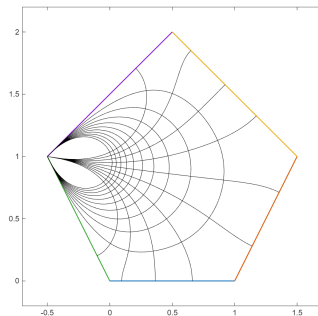


Figure 2: Map from upper half plane onto pentagon.

Plots produced using MATLAB SC-toolbox created by Tobin A. Driscoll [2].

Examples with infinite vertices

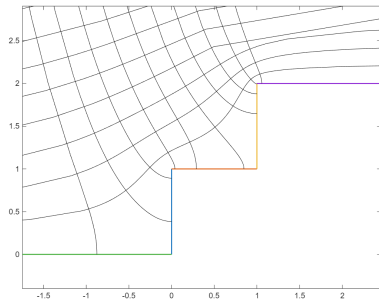


Figure 3: $f : \mathbb{H} \rightarrow$ infinity step domain.

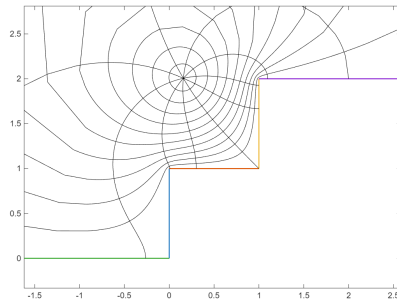


Figure 4: $f : \mathbb{D} \rightarrow$ infinity step domain.

Plots produced using MATLAB SC-toolbox created by Tobin A. Driscoll [2].

Schwarz–Christoffel Formula

The *Riemann mapping theorem* guarantees the existence of conformal maps from the unit disk onto an arbitrary simply connected domain, such as the interior of a polygon. However, it does not tell us what that map actually *is*.

The Parameter Problem: Given the vertices of a polygon, can we determine a Schwarz–Christoffel map onto the interior of that polygon?

Theorem (Driscoll, Trefethen, [4])

Let $f : \mathbb{H} \rightarrow \mathbb{P}$ be a Schwarz–Christoffel map, where $\mathbb{P}$ is a polygonal domain with vertices, $w_1, w_2, \dots, w_n$ given in counter-clockwise order, such that $f^{-1}(w_n) = \infty$. Let $\alpha_1, \alpha_2, \dots, \alpha_n$ be the corresponding interior angles. Then for the prevertices, $z_k = f^{-1}(w_k)$ for $1 \leq k \leq n-1$, we have that

$$f(z) = A + C \int^z \prod_{k=1}^{n-1} (\zeta - z_k)^{\alpha_k - 1} d\zeta$$

where A is a constant of integration and C is a scaling constant.

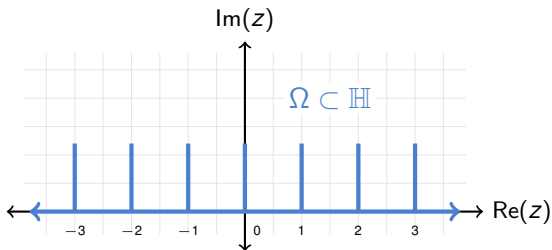
What is a comb domain?

Definition

A *comb domain* is defined to be the upper half-plane with vertical slits stemming from the real axis removed. Formally speaking, given a set of slits,

$$S_n := \{z \in \mathbb{C} : \Re z = x_n \text{ and } 0 < \Im z \leq y_n\}$$

where $x_n \in \mathbb{R}$ are the bases of the slits, $y_n > 0$ are the heights of the slits, and $n \in \mathbb{N}$, the set $\Omega = \mathbb{H} \setminus \bigcup_{n \in \mathbb{N}} S_n$ is a comb domain with slits S_n .



Geometry of comb domains

The sum of the $\alpha_k - 1$ at each slit is

$$(0.5 - 1) + (2 - 1) + (0.5 - 1) = -0.5 + 1 - 0.5 = 0$$

and since there are an integer number of slits in a period, the entire sum is 0.

Lemma

For a periodic comb domain with n slits in a period, the interior angles associated with the vertices of the slits are given by

$$\alpha_k = \begin{cases} 0.5, & \text{if } k \text{ is odd} \\ 2, & \text{if } k \text{ is even} \end{cases}$$

where $1 \leq k \leq 3n$. Furthermore, the sum of $\alpha_k - 1$ from $k = 1$ to $3n$ is

$$\sum_{k=1}^{3n} \alpha_k - 1 = \sum_{k=1}^n (-0.5 + 1 - 0.5) = 0$$

Goal

Key Questions:

- 1 Given the vertices in a period of a periodic comb, can we construct a formula for Schwarz–Christoffel maps?
- 2 Using such a formula, can we compute solutions to the parameter problem for such a comb?

Challenges:

- 1 Due to the geometry of comb domains, the base of the slits creates a singularity where the derivative of any conformal map blows up as we get closer to the vertex at the base.
- 2 **Overcrowding:** the main computational challenge which arises for Schwarz–Christoffel problem is the phenomenon when prevertices cluster together exponentially, creating unstable integrals.

Periodicity of prevertices

The periodicity of a comb domain (and its vertices) *implies* the periodicity of the prevertices under any Schwarz–Christoffel map.

Theorem

Suppose $\mathbb{P}$ is a periodic comb domain in the complex plane, with period T . Then a conformal map $f(z) : \mathbb{H} \rightarrow \mathbb{P}$ satisfies that

$$f(z + \pi) = f(z) + T$$

where the canonical period in the domain is normalized to π .

In particular, this entails that $f(z)$ is quasi-periodic, and its derivative $f'(z)$ is periodic.

Lemma

Given a comb domain, $\mathbb{P}$ and a Schwarz–Christoffel map $f : \mathbb{H} \rightarrow \mathbb{P}$, the angular derivative $f'(\infty)$ is finite, and can be normalized to 1.

The Formula

By applying the periodicity conditions we have demonstrated for SC maps onto comb domains, and using the angle-preserving properties of conformal maps we arrive at the following formula:

Theorem

Let $f : \mathbb{H} \rightarrow \mathbb{P}$ be a Schwarz–Christoffel map and $\mathbb{P}$ be a periodic comb domain with finitely many vertices per period. Then given prevertices $z_1, z_2, \dots, z_n$ in a fixed period, the map f is given by

$$f(z) = A + C \int^z \prod_{k=1}^n \left[\sin \left(\frac{\pi}{T} (\zeta - z_k) \right) \right]^{\alpha_k - 1} d\zeta$$

where A and C are complex constants, T is the period of the prevertices, and α_k are the interior angles.

Derivation of the Formula

Let $z_1, z_2, \dots, z_n$ be the prevertices corresponding to the vertices in a particular period of the comb. By Theorem 3.1, the rest of the prevertices are thus given by $z_k + jT$ for $j \in \mathbb{Z}$ and $1 \leq k \leq n$. Then

$$f'(z) = C \prod_{k=1}^n \prod_{j=-\infty}^{\infty} [z - (z_k + jT)]^{\alpha_k - 1}$$

We can rewrite the inner product as

$$\prod_{j=-\infty}^{\infty} (z - z_k - jT)^{\alpha_k - 1} = (z - z_k)^{\alpha_k - 1} \prod_{j=1}^{\infty} (-j^2 T^2)^{\alpha_k - 1} \left[1 - \frac{(z - z_k)^2}{j^2 T^2} \right]^{\alpha_k - 1}$$

Derivation of the Formula

Substituting this we get

$$\begin{aligned}
 f'(z) &= C \prod_{k=1}^n (z - z_k)^{\alpha_k - 1} \prod_{j=1}^{\infty} (-j^2 T^2)^{\alpha_k - 1} \left[1 - \frac{(z - z_k)^2}{j^2 T^2} \right]^{\alpha_k - 1} \\
 &= C \left[\prod_{j=1}^{\infty} \prod_{k=1}^n (-j^2 T^2)^{\alpha_k - 1} \right] \cdot \prod_{k=1}^n \left[(z - z_k) \prod_{j=1}^{\infty} \left(1 - \frac{(z - z_k)^2}{j^2 T^2} \right) \right]^{\alpha_k - 1}
 \end{aligned}$$

Then observe that by applying Lemma 2.4, we have that

$$\prod_{k=1}^n (-j^2 T^2)^{\alpha_k - 1} = (-j^2 T^2)^{\sum_{k=1}^n \alpha_k - 1} = (-j^2 T^2)^0 = 1$$

Derivation of the Formula

Therefore we have

$$f'(z) = C \prod_{k=1}^n \left[(z - z_k) \prod_{j=1}^{\infty} \left(1 - \frac{(z - z_k)^2}{j^2 T^2} \right) \right]^{\alpha_k - 1}$$

Furthermore, by the Weierstrass factorization theorem, we have that

$$\sin(\pi z) = \pi z \prod_{j=1}^{\infty} \left[1 - \frac{z^2}{j^2} \right]$$

Using this, we get

$$\begin{aligned} \prod_{j=-\infty}^{\infty} (z - z_k - jT) &= (z - z_k) \prod_{j=1}^{\infty} \left[1 - \frac{(z - z_k)^2}{j^2 T^2} \right] \\ &= \frac{T}{\pi} \sin \left(\frac{\pi}{T} (z - z_k) \right) \end{aligned}$$

Derivation of the Formula

Therefore, we have that

$$f'(z) = C \prod_{k=1}^n \left[\frac{T}{\pi} \sin \left(\frac{\pi}{T} (z - z_k) \right) \right]^{\alpha_k - 1}$$

and using $\prod_{k=1}^n \left(\frac{T}{\pi} \right)^{\alpha_k - 1} = 1$ and integrating we get

$$f(z) = A + C \int^z \prod_{k=1}^n \left[\sin \left(\frac{\pi}{T} (\zeta - z_k) \right) \right]^{\alpha_k - 1} d\zeta$$

where A is the integration constant.

The Reformulation

In this formulation, the $\sin$ term behaves as

$$\sin(w_k) = \frac{e^{iw_k} - e^{-iw_k}}{2i} = \frac{-e^{-iw_k}}{2i}(1 - e^{2iw_k})$$

where $w_k(z) := \frac{\pi}{T}(z - z_k)$. As $z \rightarrow \infty$ in the upper half-plane, $\sin(w_k(z)) \rightarrow \infty$. Thus, we need a reformulation:

Corollary

For a given Schwarz–Christoffel map $f : \mathbb{H} \rightarrow \mathbb{P}$ where $\mathbb{P}$ is a comb domain, the derivative $f'(z)$ can be reformulated as:

$$f'(z) = C \prod_{k=1}^n \left[1 - e^{2i \frac{\pi}{T}(z - z_k)} \right]^{\alpha_k - 1}$$

Derivation of the Reformulation

Given that

$$\sin(w_k) = \frac{e^{iw_k} - e^{-iw_k}}{2i} = \frac{-e^{-iw_k}}{2i}(1 - e^{2iw_k})$$

where $w_k := \frac{\pi}{T}(z - z_k)$, the derivative of the SC map, $f(z)$ is

$$\begin{aligned} f'(z) &= C \prod_{k=1}^n \left[\frac{-e^{-iw_k}}{2i} (1 - e^{2iw_k}) \right]^{\alpha_k - 1} \\ &= C \cdot \prod_{k=1}^n \left(\frac{-1}{2i} \right)^{\alpha_k - 1} \cdot \prod_{k=1}^n (e^{-iw_k})^{\alpha_k - 1} \prod_{k=1}^n (1 - e^{2iw_k})^{\alpha_k - 1} \end{aligned}$$

The constant term is

$$\prod_{k=1}^n \left(\frac{-1}{2i} \right)^{\alpha_k - 1} = \left(\frac{-1}{2i} \right)^{\sum_{k=1}^n \alpha_k - 1} = \left(\frac{-1}{2i} \right)^0 = 1$$

Derivation of the Reformulation

Additionally, we have that the second term can be rewritten as

$$\prod_{k=1}^n (e^{-iw_k})^{\alpha_k - 1} = \exp \left[-i \frac{\pi}{T} z \sum_{k=1}^n (\alpha_k - 1) + i \frac{\pi}{T} \sum_{k=1}^n z_k (\alpha_k - 1) \right]$$

Recall that the sum of $\alpha_k - 1$ over a period is 0. Using this geometric property, we get

$$\prod_{k=1}^n (e^{-iw_k})^{\alpha_k - 1} = \exp \left[i \frac{\pi}{T} \sum_{k=1}^n z_k (\alpha_k - 1) \right]$$

Since this term is not dependent on z , it is also constant, and thus can be absorbed into our scaling factor, C , leaving us with

$$f'(z) = C \prod_{k=1}^n \left(1 - e^{2i \frac{\pi}{T} (z - z_k)} \right)^{\alpha_k - 1}$$

The Singularities of the Slits

To solve the parameter problem, we must compute the physical boundary lengths from our current guess for the prevertices.

The distance between vertices w_k and w_{k+1} should correspond with the image of the interval $[x_k, x_{k+1}]$, that is

$$L_{k,k+1} = \int_{x_k}^{x_{k+1}} |f'(x)| dx = \int_{x_k}^{x_{k+1}} C \prod_{j=1}^{3n} |\sin(x - x_j)|^{\alpha_j - 1} dx$$

However, the integrand has a singularity at the base of a slit, since the product has a factor of

$$|\sin(x_j - x_j)|^{\alpha_j - 1} = |\sin(0)|^{-0.5} = \frac{1}{\sqrt{0}} = \infty$$

Extracting the Singularities

To handle this singularity, we can extract it as follows;

$$|\sin(x - x_j)|^{\alpha_j - 1} = |x - x_j|^{\alpha_j - 1} \cdot \left(\frac{|\sin(x - x_j)|}{|x - x_j|} \right)^{\alpha_j - 1}$$

Observe that the term on the right satisfies

$$\lim_{x \rightarrow x_j} \left(\frac{|\sin(x - x_j)|}{|x - x_j|} \right)^{\alpha_j - 1} = 1$$

and so this term remains smooth and bounded at each prevertex. Then we can define the function,

$$g(x) := C \left(\frac{|\sin(x - x_k)|}{|x - x_k|} \right)^{\alpha_k - 1} \left(\frac{|\sin(x - x_{k+1})|}{|x - x_{k+1}|} \right)^{\alpha_{k+1} - 1} \prod_{j \neq k, k+1} |\sin(x - x_j)|^{\alpha_j - 1}$$

which is smooth along the interval $[x_k, x_{k+1}]$. Then we can write the modulus of the derivative of f as

$$|f'(x)| = |x - x_k|^{\alpha_k - 1} |x - x_{k+1}|^{\alpha_{k+1} - 1} |g(x)|$$

Applying Gauss-Jacobi Quadrature

To apply Gauss–Jacobi quadrature to estimate $L_{k,k+1}$ we need to translate the interval $[x_k, x_{k+1}]$ to the canonical interval $[-1, 1]$ by applying the substitution;

$$x(u) := \frac{x_{k+1} - x_k}{2}u + \frac{x_k + x_{k+1}}{2}$$

which gives the length to be

$$L_{k,k+1} = \left(\frac{x_{k+1} - x_k}{2} \right)^{\alpha_k + \alpha_{k+1} - 1} \int_{-1}^1 (1-u)^{\alpha_{k+1}-1} (1+u)^{\alpha_k-1} g(x(u)) du$$

Since $\alpha_k - 1$ and $\alpha_{k+1} - 1$ are either -0.5 or 1 , which are both greater than -1 , we can apply Gauss–Jacobi quadrature to estimate this integral.

Residual Optimization Problem

For a given interval guess $\vec{p}$, we compute the integrated slit heights $\vec{h}(\vec{p})$ and horizontal spacings $\vec{s}(\vec{p})$ using Gauss–Jacobi quadrature. To handle the scaling constant C , we dynamically calibrate it at each iteration such that

$$C(\vec{p}) = \frac{W}{\sum_{i=1}^n s_i(\vec{p})}$$

Let $\vec{H}$ and $\vec{S}$ represent the known target heights and target spacings of the comb domain, respectively. Then we can evaluate our guess using the residual vector function $F(\vec{p})$ defined as:

$$F(\vec{p}) := \begin{bmatrix} C(\vec{p}) \cdot \vec{h}(\vec{p}) - \vec{H} \\ C(\vec{p}) \cdot \vec{s}(\vec{p}) - \vec{S} \end{bmatrix}$$

To solve the system $F(\vec{p}) = \vec{0}$, we employ a bounded nonlinear least-squares optimization routine. The system is solved numerically using the Trust Region Reflective algorithm to minimize $\|F(\vec{p})\|^2$. [1]

Plot resolution

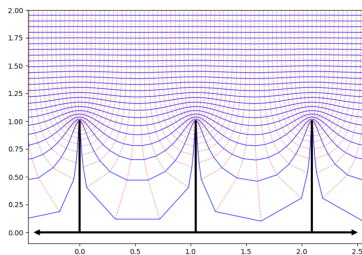


Figure 5: Uniform comb domain.

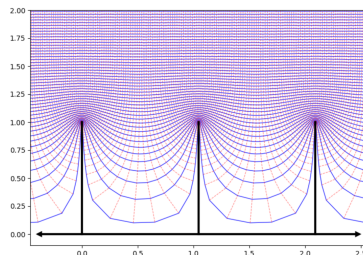


Figure 6: Effect of increasing resolution.

Increasing number of periods displayed

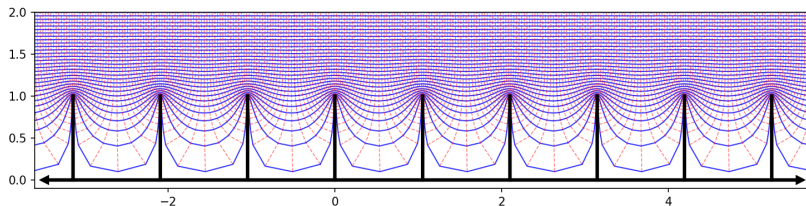


Figure 7: Uniform comb domain with 3 periods displayed.

Lowering the floor

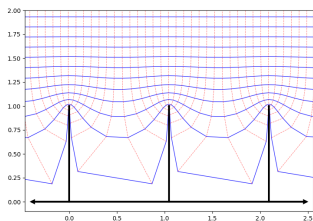


Figure 8: $\varepsilon = 0.01$

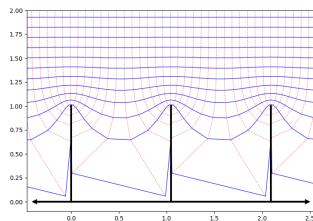


Figure 9: $\varepsilon = 0.001$

Lowering the floor

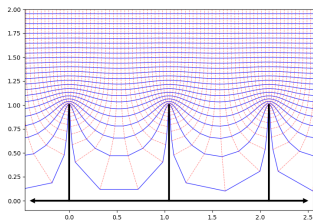


Figure 10: $\varepsilon = 0.01$

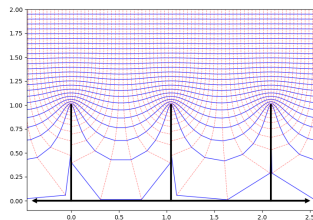


Figure 11: $\varepsilon = 0.001$

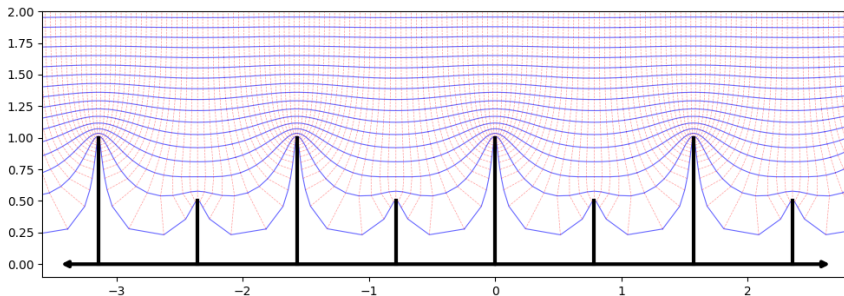


Figure 12: Alternating slit heights with two slits per period, four periods shown.

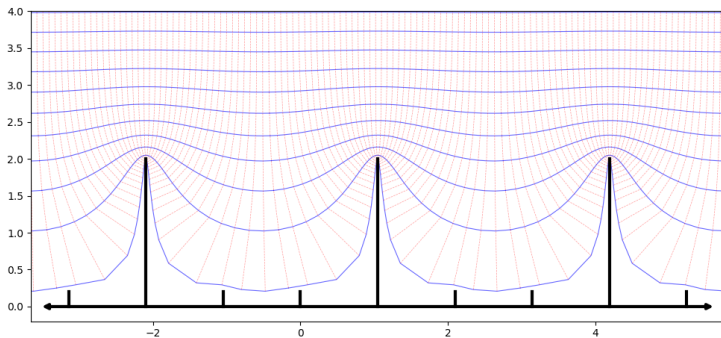


Figure 13: Comb domain with large variance between slit heights.

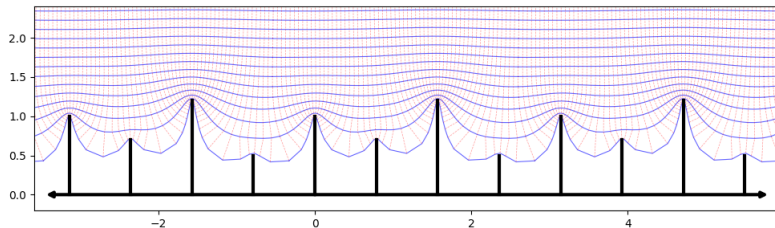


Figure 14: Comb domain with large variance between slit heights.

References

- [1] Mary Ann Branch, Thomas F. Coleman, and Yuying Li. A subspace, interior, and conjugate gradient method for large-scale bound-constrained minimization problems. *SIAM Journal on Scientific Computing*, 21, 1999.
- [2] Tobin A. Driscoll. A MATLAB toolbox for Schwarz–Christoffel mapping. *ACM Transactions on Mathematical Software (TOMS)*, 1996.
- [3] Christian Pommerenke. *Boundary Behaviour of Conformal Maps*. Springer-Verlag Berlin Heidelberg, 1992.
- [4] Lloyd N. Trefethen Tobin A. Driscoll. *Schwarz–Christoffel Mapping*. Cambridge University Press, 2002.

Angular derivative

In order to derive a new Schwarz–Christoffel formula for comb domains, we must ensure that the angular derivative does not blow up as $z \rightarrow \infty$ in the upper half-plane.

Theorem

Given a comb domain, $\mathbb{P}$ and a Schwarz–Christoffel map $f : \mathbb{H} \rightarrow \mathbb{P}$, the angular derivative $f'(\infty)$ is finite, and can be normalized to 1.

Lemma (Pommerenke, [3])

Let $f_j : \mathbb{H} \rightarrow G_j$ be conformal maps onto domains G_1 and G_2 , where $G_1 \subset G_2$. Suppose $\Gamma \subset G_1$ is a Jordan curve ending at some $w \in \partial G_1 \cap \partial G_2$, and consider the preimages, $\Gamma_j = f_j^{-1}(\Gamma)$, with ends $z_j \in \partial \mathbb{H}$. If the angular derivative $f_1'(z_1)$ exists and is finite then $f_2'(z_2)$ likewise exists and is finite.

Stability of the reformulation

Let $z = x + iy$ where $x, y \in \mathbb{R}$, and observe that if we let $A := \frac{2\pi}{T} \in \mathbb{R}$, then we have that

$$e^{2i\frac{\pi}{T}(z-z_k)} = e^{iA(x+iy-z_k)} = e^{A(ix-y-z_k)}$$

and furthermore that

$$|e^{2i\frac{\pi}{T}(z-z_k)}| = |e^{A(ix-y-z_k)}| = |e^{iAx}| \cdot |e^{-Ay}| \cdot |e^{-Az_k}|$$

Here, we have that $|e^{iAx}| = 1$, and $|e^{-Az_k}|$ is simply constant, while $|e^{-Ay}| \rightarrow 0$ as $y \rightarrow +\infty$. In particular, since we are operating on the upper half-plane, $\mathbb{H} = \{x + iy : y > 0\}$, we know that y must be positive. Thus, this version of the derivative remains numerically stable.

Integrating around the slits

To generate a plot of the Schwarz–Christoffel map $f(z)$ onto a periodic comb domain, we need to evaluate f on the interior points of $\mathbb{H}$. This requires evaluating the map numerically using standard quadrature. That is, for some $z \in \mathbb{H}$, we need to estimate the definite integral:

$$f(z) = w_k + \int_{x_k}^z f'(\zeta) d\zeta$$

where (x_k, w_k) is an arbitrary vertex-prevertex pair which we fix to establish the spatial integration constant, w_k .

However, if x_k corresponds to a base of a slit, the derivative $f'(\zeta)$ blows up, creating an integrable singularity at the anchor. Specifically, because the interior angle at a base is $\pi/2$, we have $\alpha_k = 0.5$. Thus, $f'(\zeta) \sim \mathcal{O}((\zeta - x_k)^{-0.5})$. As the integration path approaches the prevertex ($\zeta \rightarrow x_k$), the integrand $f'(\zeta) \rightarrow \infty$.

Change of variables

In order to handle this boundary singularity numerically, we parameterize the straight-line path from x_k to z using a nonlinear change of variables:

$$\zeta(t) = x_k + (z - x_k) \sin^2 \left(\frac{\pi}{2} t \right), \quad t \in [0, 1]$$

The differential element along this path is:

$$d\zeta = (z - x_k) \pi \sin \left(\frac{\pi}{2} t \right) \cos \left(\frac{\pi}{2} t \right) dt$$

For values of t near 0 (where the path originates at the singularity x_k), we can approximate $\sin(\frac{\pi}{2} t)$ by $\frac{\pi}{2} t$. Thus, the distance from the singularity behaves as $\zeta(t) - x_k \sim \mathcal{O}(t^2)$.

Stabilization of the Integral

Substituting this into our asymptotic derivative, we have for small t :

$$f'(\zeta(t)) \sim \mathcal{O}((t^2)^{-0.5}) = \mathcal{O}(t^{-1})$$

Furthermore, since the Jacobian behaves as $\sin(\frac{\pi}{2}t) \cos(\frac{\pi}{2}t) \approx t \cdot 1$ for small t , we have that $d\zeta \sim \mathcal{O}(t)dt$. Therefore, the full integrand evaluates to:

$$f'(\zeta(t)) \frac{d\zeta}{dt} \sim \mathcal{O}(t^{-1}) \cdot \mathcal{O}(t) = \mathcal{O}(1)$$

This substitution allows us to handle the singularities at the base of slits, yielding a strictly bounded and continuous integrand that is numerically stable for standard adaptive quadrature. We can use this to then evaluate the map within the canonical period $[0, \pi]$, and then exploit the periodicity of the comb domain to continue the plot to the next period by shifting by the period of the comb.

Proof of Theorem 3.1

Let $f : \mathbb{H} \rightarrow \mathbb{P} \subset \hat{\mathbb{C}}$ be a conformal map onto a periodic region, $\mathbb{P}$ in the complex plane, and oriented so that $f(\infty) = \infty \in \partial\mathbb{P}$.

Then the map;

$$g(z) := f^{-1}(f(z) + T) : \mathbb{H} \rightarrow \mathbb{H}$$

is an automorphism on the upper half-plane. Since the upper half-plane is conformally equivalent to the unit disk by Theorem 2.2, and so $g(z)$ has the form

$$g(z) = \frac{az + b}{cz + d} \quad \text{for} \quad \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in PSL_2(\mathbb{R})$$

i.e. a 2×2 real-valued matrix with determinant 1.

Proof of Theorem 3.1

Thus, we have that

$$\frac{az + b}{cz + d} = f^{-1}(f(z) + T) \implies f\left(\frac{az + b}{cz + d}\right) = f(z) + T$$

Since $f(\infty) = \infty$, we must have that

$$\infty = \frac{a \cdot \infty + b}{c \cdot \infty + d} = \begin{cases} \frac{a}{c}, & \text{if } c \neq 0 \\ \infty, & \text{if } c = 0 \end{cases}$$

Since this value is ∞ , we must have $c = 0$, and so $f\left(\frac{az+b}{d}\right) = f(z) + T$. Furthermore, since $1 = ad - bc = ad$, we must have $d = \frac{1}{a}$, and so we can rewrite this as

$$f(a^2z + ab) = f(z) + T$$

Proof of Theorem 3.1

Since the angular derivative of f at ∞ is finite, we can normalize it so that

$\lim_{z \rightarrow \infty} \frac{f(z)}{z} = 1$. Thus, we have that in the limit at infinity

$$1 = \lim_{z \rightarrow \infty} \frac{f(a^2 z + ab)}{a^2 z + ab} = \lim_{z \rightarrow \infty} \frac{f(z) + T}{a^2 z + ab} = \lim_{z \rightarrow \infty} \frac{\frac{f(z)}{z} + \frac{T}{z}}{a^2 + \frac{ab}{z}}$$

Since $\frac{T}{z}, \frac{ab}{z} \rightarrow 0$, we are left with

$$1 = \frac{\lim_{z \rightarrow \infty} \frac{f(z)}{z}}{a^2} = \frac{1}{a^2} \implies a^2 = 1 \implies a = 1$$

Therefore, f reduces to a translation of the form $f(z + ab) = f(z) + T$, where ab is simply a constant, which we can normalize to be π by convention. Differentiating this, we clearly get that

$$f'(z + \pi) = f'(z) + \frac{d}{dz} T = f'(z)$$

since T is a constant, and so the derivative is periodic.